

Question				Marking Details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	i		60 000 : 1 Accept correct ratio not to 1 e.g. 120 000:2		1		1	1	
		ii		<i>Advantage:</i> Any one from: Reduces water loss/ allows them to live in arid conditions (1) No {blood / circulatory system/ pigment} required (1) Oxygen supplied directly to the cells/ muscles (1) Tracheoles go directly into cells/ tissues (1) <i>Disadvantage:</i> Size/ shape limitation (1)	2			2		
	(b)			Any 3 (x1) from: • Diffusion of gases related to Surface Area (1) • Oxygen use related to volume (1) • Amoeba has large Surface Area : volume ratio (1) • Therefore diffusion of gases sufficient (to supply demand) (1) • Short diffusion distance (1)	3			3		
	(c)			Ventilation (movements/system)/ description of replacing oxygen in alveoli/ ORA for carbon dioxide (1) {Blood/ transport system} (takes oxygen away from respiratory surface)/ ORA for carbon dioxide/ or description of (1)	2			2		

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	(d)		Any two (x1) from: - Low body temperature (1) - Metabolic rate low / example of (metabolic) reactions not required/ working at lower rate(1) - Less energy required for support / fish are buoyant (1) ORA for dog Reject dogs move more than fish/ dogs are more active than fish/ more oxygen needed for respiration		2		2		
	(e)		1. Mammalian (red blood cells) evolved when O ₂ levels were lower/ ORA (1) 2. {Biconcave/shape} increases surface area (for increased O ₂ absorption) (1) 3. No nucleus + can carry more haemoglobin (so increased O ₂ transport) / ORA(1) 4. Thin centres/small/ biconcave so short diffusion distance (so faster diffusion) (1) 5. Small(er) in size so have a higher sa : vol ratio (1) 6. Small(er), {so more of them/ total surface area larger} (1) Accept ref. to smaller capillaries qualified/ biconcave shape gives flexibility to fit through capillaries		4		4		
			Question 1 total	7	7	0	14	1	0